

Amar Dhillon

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EDUCATION

University of Maryland, College Park
B.S. in Computer Science, Minor in Statistics; GPA: 3.5

College Park, MD
May 2028

CERTIFICATIONS

Amazon Web Services (AWS) Certified Developer - Associate
Amazon Web Services

Nov 2024 – Nov 2027
Digital Badge

EXPERIENCE

Software Development Engineer Intern
Amazon Web Services

Sep 2026 – Dec 2026
Redmond, WA

- Researching and implementing Java-based utilization optimizations for Amazon Redshift

Software Engineering Intern
Capital One

Jun 2026 – Aug 2026
Richmond, VA

- Overhauled the legacy UI of a test data workflow creator using React + Electron with TypeScript workflow validation backed by Jest testing and a React-Flow editor, receiving **90%** satisfaction compared to the legacy tool
- Engineered a precomputed tree-style workflow search index to minimize database queries under access constraints, distilling popular workflows from a **270K+** row PostgreSQL table with Python + SQLAlchemy, backed by Redis caching
- Developed a Claude skill for workflow generation using sub-skill decomposition and Python functions for workflow search and validation, guided by CloudWatch + OpenSearch latency logs, cutting generation time **65%** to under 15 seconds

Software Engineer (Contract)
Amazon (via UMD App Development Contracting)

Jan 2026 – Present
College Park, MD

- Deployed an Amazon Leo space topic research utility adopted into Leo's ecosystem through Docker + AWS Fargate and GitHub Actions CI/CD, with AWS CDK (TypeScript) infrastructure so new engineers can contribute
- Achieved a **34%** speedup on a pipeline routing 6,000+ scraped research documents for inference by implementing Kinesis streams feeding Lambda processors, with API Gateway fronting for app integration
- Integrated automated fine-tuning of an AWS Bedrock source relevancy model through EventBridge cron triggers with decision gates to avoid redundant retraining, with X-Ray tracing and SNS failure alerts

Machine Learning Research Intern
UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present
College Park, MD

- Developing automated training for PhysTwin, a physics-based digital twin, with a PhD research team targeting an ICLR submission
- Built a Python pipeline linking AnyGrasp multi-camera grasp candidates to Open3D overlay views consumed by a Qwen3-VL-8B VLM, enabling researchers to isolate scene components
- Cut research-environment setup time **30%** via uv dependency pinning and Docker, with psutil timing and memory monitoring and Pytest unit testing for identifying bottlenecks

Co-Founder, Software Engineer
TerpLabs

Jun 2025 – Present
College Park, MD

- Co-founded and scaled a **110+** member campus product studio shipping 5 live products, organizing teams through agile practices
- Sustained **20,000+** visits in the first 30 days of Tortuga, a class scheduler, through developing Prisma-based SQL querying with Redis query caching to reduce database load and latency
- Building a RAG meal suggestion feature for the official UMD app using GPT-4o mini over ChromaDB + SentenceBERT vector search on scraped recipes, achieving **93%** meal satisfaction on internal test users

PROJECTS

IEX Matching Engine Simulation | C++, STL, POSIX

Jul 2026 – Aug 2026

- Implemented a C++ order book simulation on historical IEX data, with a contract-based plug-and-play design for evaluating multiple approaches with enforced correctness
- Wrote paginated hash-based storage for live order book data with in-memory LFU caching for frequently requested tickers, evaluated for drift against IEX's historical book data
- Designed and benchmarked a time-based adaptive LSM storage against B+ tree storage for simulation logs, testing throughput and latency via rdtsc timing and hardware utilization via Linux perf

TECHNICAL SKILLS

Languages: Python, Java, C++, C, TypeScript/JavaScript, SQL

AWS: Lambda, Kinesis, Fargate/ECS, API Gateway, EventBridge, Bedrock, CDK, CloudWatch, X-Ray, SNS, IAM

Backend & Data: Node.js, React, Electron, PostgreSQL, Redis, SQLAlchemy, Prisma, OpenSearch, ChromaDB

AI/ML & Tools: PyTorch, Hugging Face, RAG, Open3D, Docker, GitHub Actions, Linux, Pytest, Jest, uv, perf